



Time restricted eating helps

According to recent study, time-restricted eating can turn out to be beneficial for blood glucose.
<https://digit.in/may19-61>



Children should play more

New WHO guidelines on physical activity, sedentary behaviour and sleep for children.
<https://digit.in/may19-62>

nature and low degree of complexity. However, a number of medications and drugs cannot be administered this way since the presence of certain chemicals make it impossible for them to survive passage through the stomach or gastrointestinal tract. They either rapidly degrade or lead to poor absorption in the gastrointestinal tract when administered orally. Thus, any system looking to facilitate oral delivery of drugs like insulin has to figure out a way to make them survive rapid degradation and also a way to administer them where they are most effective. Researchers from MIT might have found the most effective way to do so. However, they haven't been the first to come this close.

PAST WORK

This is not the first time that oral delivery of insulin has been attempted. Last year, researchers from the Harvard John A. Paulson School of Engineering and Applied Sciences in Cambridge, MA, developed a new coating that was strong enough to survive the acidic environment of the stomach and dissolve only in the small intestine. They achieved this by dispersing insulin in a two-part liquid: a nutrient called choline, and a substance called geranic acid that is used as a food flavouring agent and is found naturally in cardamom. This method worked quite well. However, these capsules aren't expected anytime soon either, since there's a lot of testing left.

Their success is attributed to them being able to do three things: the liquid in which the insulin was dispersed inside the capsule prevents the hormone from being broken down by enzymes in the digestive system after the capsule dissolves, helps the insulin to pass through the mucus layer of the intestines, and opens up the seal between adjacent cells lining the intestines, helping the insulin to pass into the blood vessels.

Another added benefit of this method has been that the liquid remains stable for two weeks at room temperature and four months if refrigerated.

In comparison, injection-based insulin must be refrigerated and still doesn't last more than a few weeks.

A NEW PILL

Recent developments from MIT indicate that a team of researchers have taken a big step towards oral delivery of insulin. They have developed a 'blueberry-sized' capsule filled with compressed insulin that has shown good results in the lab for oral delivery. The design comprises of a biodegradable capsule which contains an insulin microneedle. Upon swallowing, the insulin gets directly injected into the stomach wall. Since the stomach wall is devoid of pain receptors, researchers expect this method to be pain-free.



The MIT Insulin Pill

For those unaware, microneedles are specifically developed to be pain-free. At the millimetre scale, they are capable of creating very small perforations when penetrating the skin. For this particular research study, the microneedle was comprised of two parts: a tip comprising compressed insulin, which penetrates the stomach wall, and a biodegradable shaft, which holds the tip in place.

That's not it, in terms of design complexity. Inside the capsule, there's a sugar disc and a compressed spring that the needle is attached to. When the capsule enters the stomach, the sugar dissolves and releases the compressed spring, which, in turn, pushes the needle out and injects the insulin into the stomach wall. You might ask what is stopping the needle from firing incorrectly either prematurely or in the wrong direction. This is where the

researchers have taken inspiration from nature.

The inspiration to solve the alignment problem comes from a species of tortoise indigenous to Eastern and Southern Africa. The Leopard turtle has a shell that is high, steep and domed, which allows it to pivot back onto its feet when it is overturned. Using this design method, researchers designed the capsule to be able to self-correct its orientation and always deliver the insulin to the stomach wall. According to Alex Abramson, a graduate student at MIT and first author of the study, even if the person who has consumed the capsule is moving around, or experiences stomach growling at the point of injection, the capsule would not move from its preferred orientation.

Once the injection has been administered, the insulin enters the bloodstream and the microneedle tip dissolves, which it has been designed to do. The capsule itself passes through the digestive system without any side effects. While it took about one hour in the particular study for the entire process, the researchers have established that the rate can be controlled to some extent by the way the microneedle is made. Doses of up to 5 milligrams can be administered using this method. The team is currently working on improving the efficiency of the pill itself. The device is not expected to hit pharmacies any time soon.

CONCLUSION

As stated earlier, Diabetes is India's fastest growing disease. India has even been called the diabetes capital of the world. Simplifying the means to access necessary medication for diabetes means that anyone who is capable of taking a capsule can avail the benefits of insulin. On top of that, a medication that does not require immediate refrigeration makes itself accessible to a greater part of the affected population. These developments are not ready for the market yet, but when they are, they will benefit countries like ours the most. 